

DIGITAL IMAGE PROCESSING

LECTURE # 5

IMAGE ENHANCEMENT IN SPATIAL DOMAIN-II

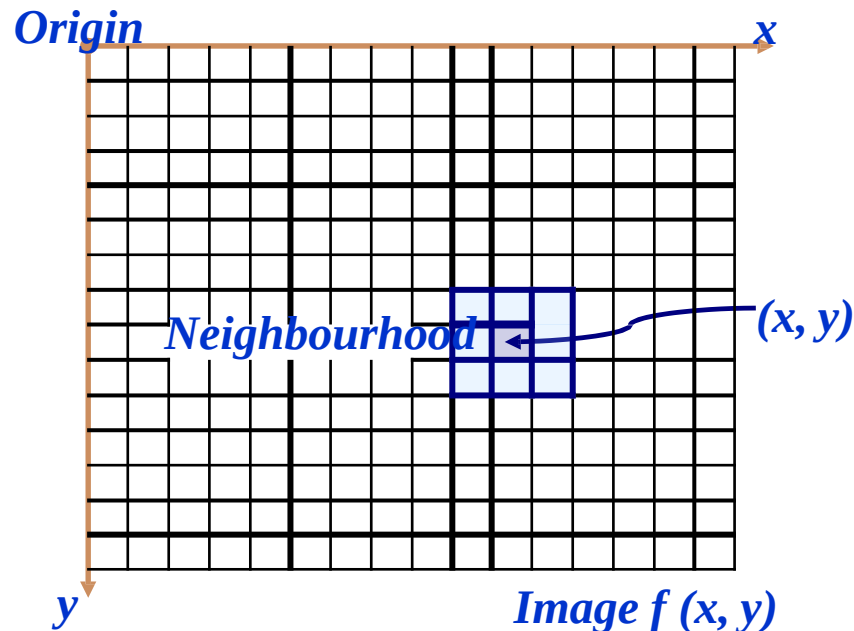
Topics to Cover

- Local/ Neighborhood Operations on images
- Local Enhancement through Spatial Filtering
- Smoothing Spatial Filtering
 - ❑ Linear Filters
 - ❑ Non-Linear Filters
- Linear Smoothing Spatial Filtering
 - ❑ Average Filters
 - ❑ Weighted Average Filters
- Non-Linear Smoothing Spatial Filtering
 - ❑ Median Filters
 - ❑ Minimum Filters
 - ❑ Maximum Filters
- Border Pixel Treatment
- Image Histogram
- Histogram Equalization

Neighborhood Operations on Images

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- Neighbourhood operations simply operate on a larger neighbourhood of pixels than point operations
- Neighbourhoods are mostly selected as square regions around a central pixel
- Any size and any shape filter are possible, but usually odd number filter size is preferred



Local Operations through Spatial Filtering

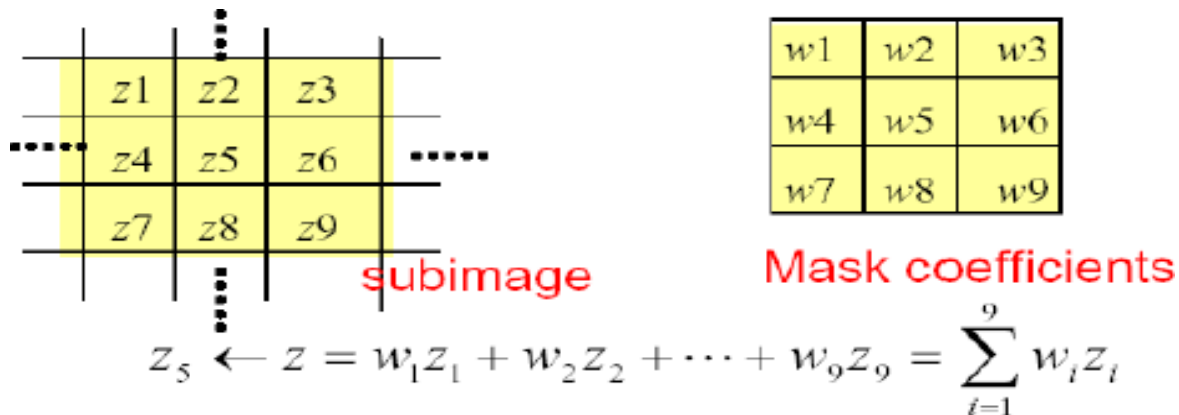
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- ❑ The output intensity value at (x,y) depends not only on the input intensity value at (x,y) but also on the specified number of neighboring intensity values around (x,y)
- ❑ **Spatial masks** (also called window, filter, kernel, template) are used and **convolved over the entire image for local enhancement** (spatial filtering)
- ❑ The size of the mask determines the number of neighboring pixels which influence the output value at (x,y)
- ❑ The values (coefficients) of the mask determine the nature and properties of enhancing technique

Basics of Spatial Filtering

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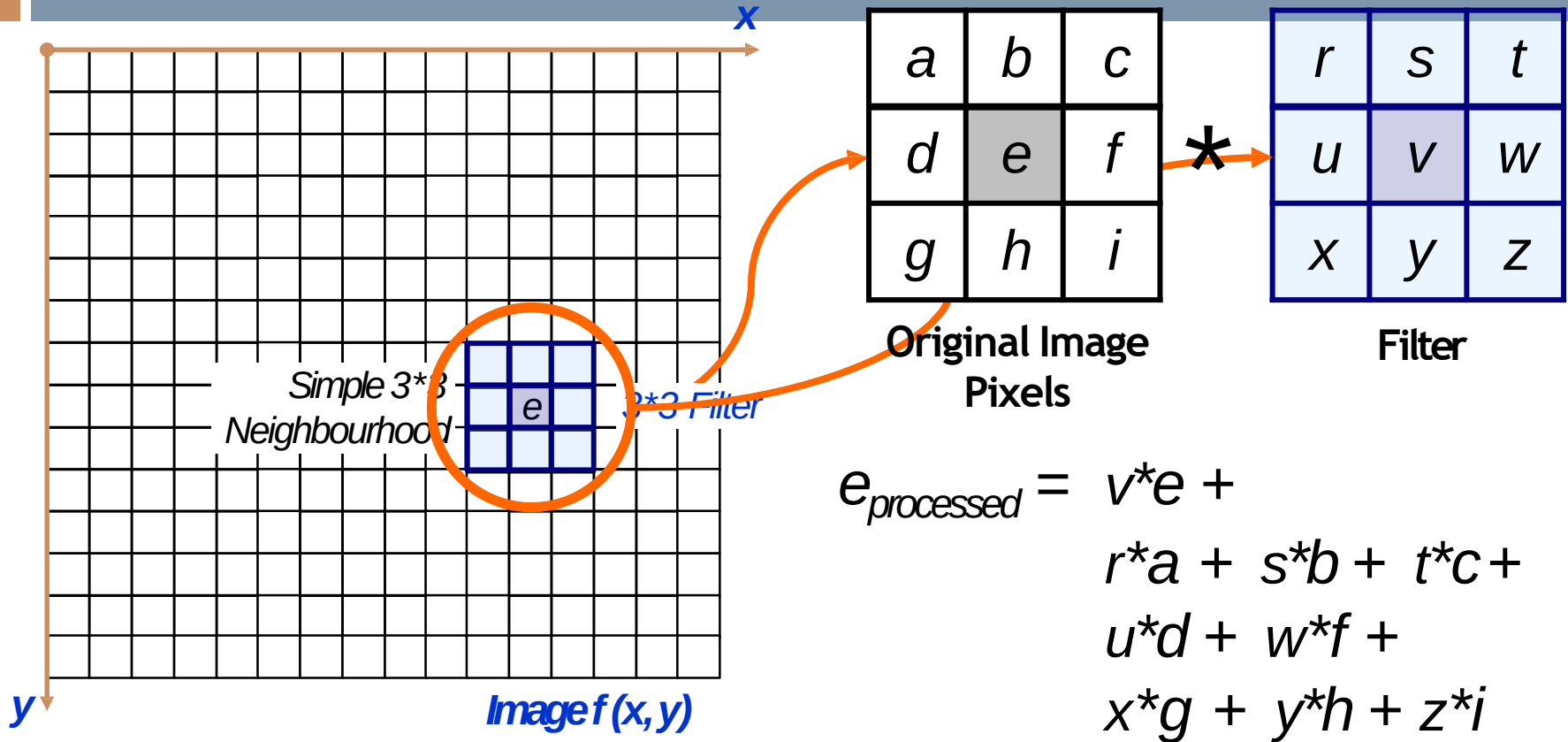
- Given the 3×3 mask with coefficients: $w_1, w_2, \dots, w_9$
- The mask covers the pixels with gray levels: $z_1, z_2, \dots, z_9$



- z gives the output intensity value for the processed image (to be stored in a new array) at the location of z_5 in the input image

Local Operations through Spatial Filtering

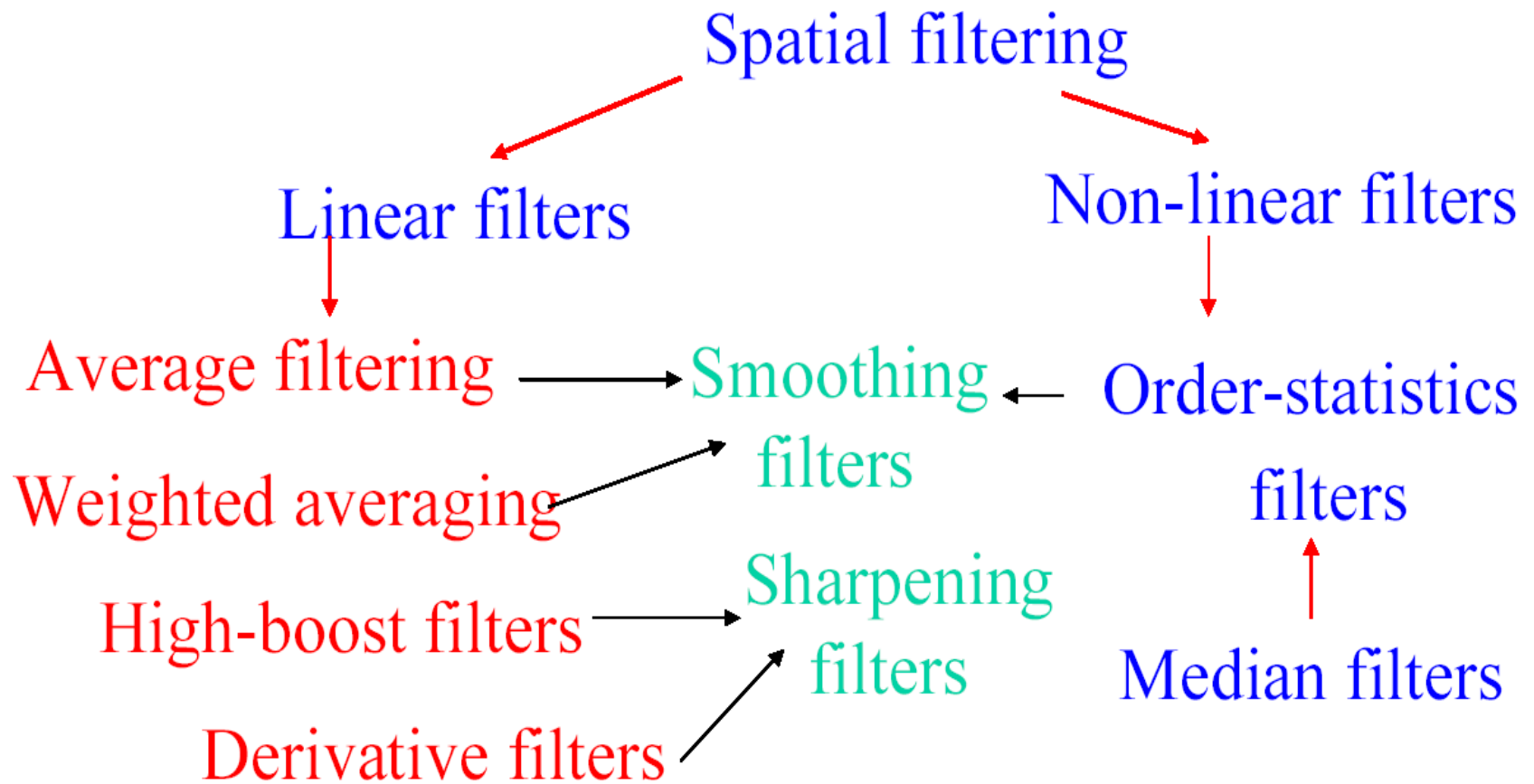
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- The above is repeated for every pixel in the original image to generate the smoothed image

Types of Spatial Filtering

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Spatial Filtering for Smoothing

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- ❑ For blurring/noise reduction;
- ❑ Blurring is usually used in preprocessing steps, e.g., to remove small details from an image prior to object extraction, or to bridge small gaps in lines or curves
- ❑ **Equivalent to Low-pass spatial filtering in frequency domain** because smaller (high frequency) details are removed based on **neighborhood averaging** (averaging filters)
- ❑ **Implementation:**
 - ✓ The simplest form of the spatial filter for averaging is a square mask (assume $m \times m$ mask) with the same coefficients $1/m^2$ to preserve the gray levels (**averaging**).
- ❑ **Applications:** Reduce noise; smooth false contours
- ❑ **Side effect:** Edge blurring

Smoothing Spatial Filters

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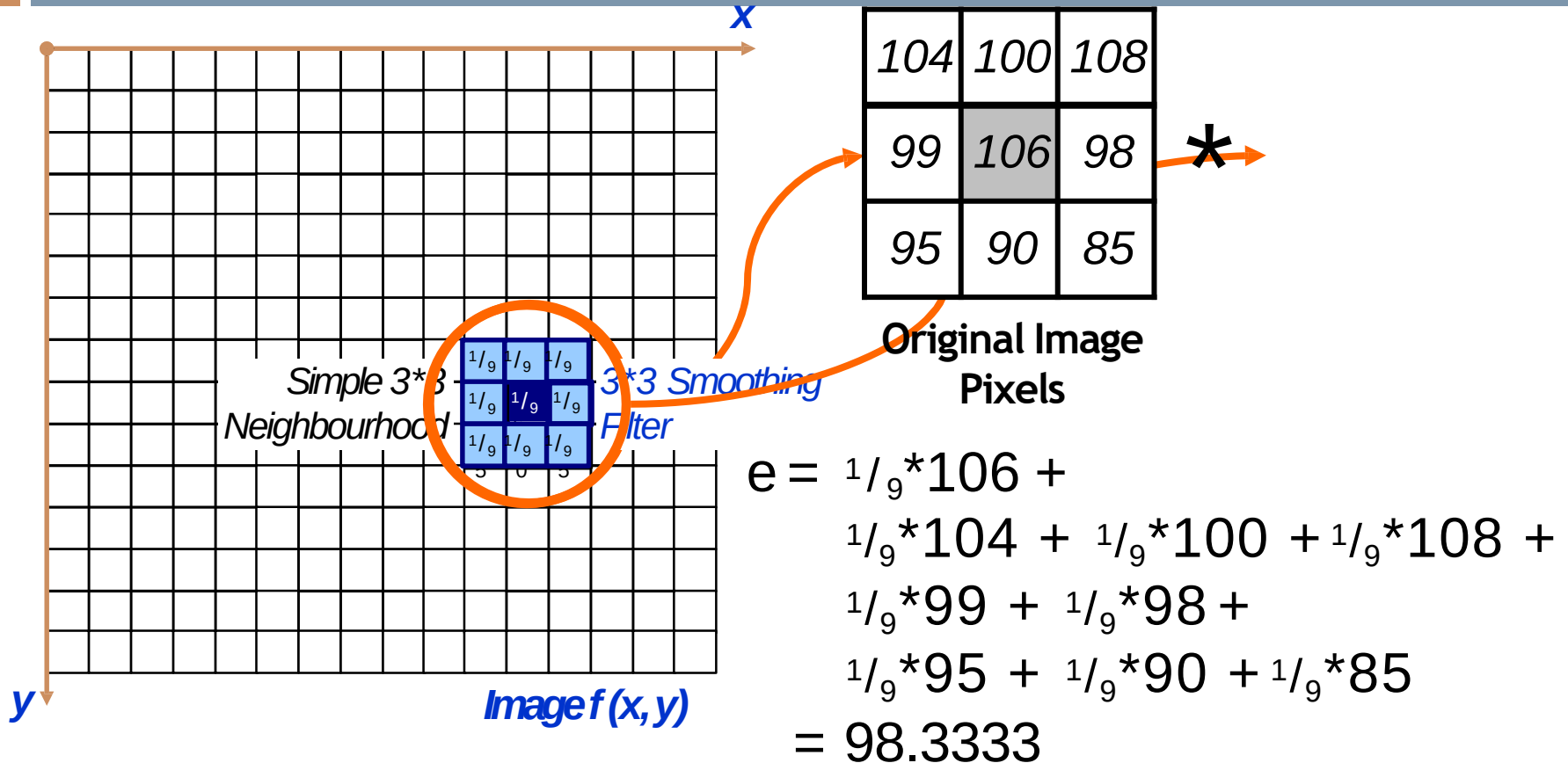
- One of the simplest **spatial filtering** operations we can perform is a smoothing operation
- **Simply average** all of the pixels in a neighbourhood around a central value
- Especially useful in removing noise from images

$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$

Simple Averaging Filter

Smoothing Spatial Filters

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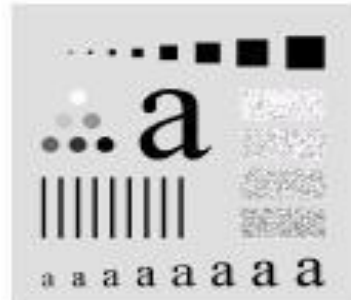


- The above is repeated for every pixel in the original image to generate the smoothed image

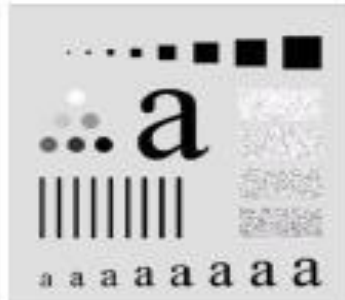
Spatial Filtering for Smoothing:: Example

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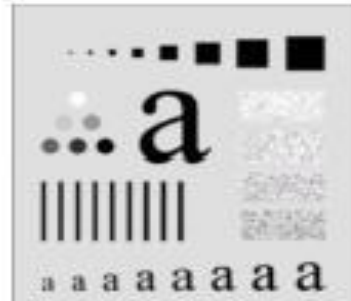
Original image
size: 500 x 500



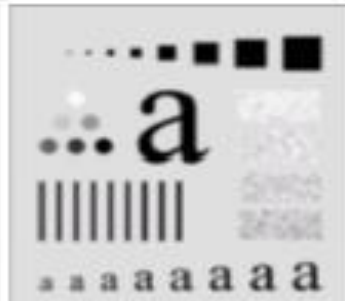
Smoothed by
3 x 3 box filter



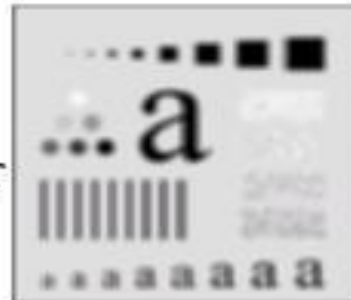
Smoothed by
5 x 5 box filter



Smoothed by
9 x 9 box filter



Smoothed by
15 x 15 box filter

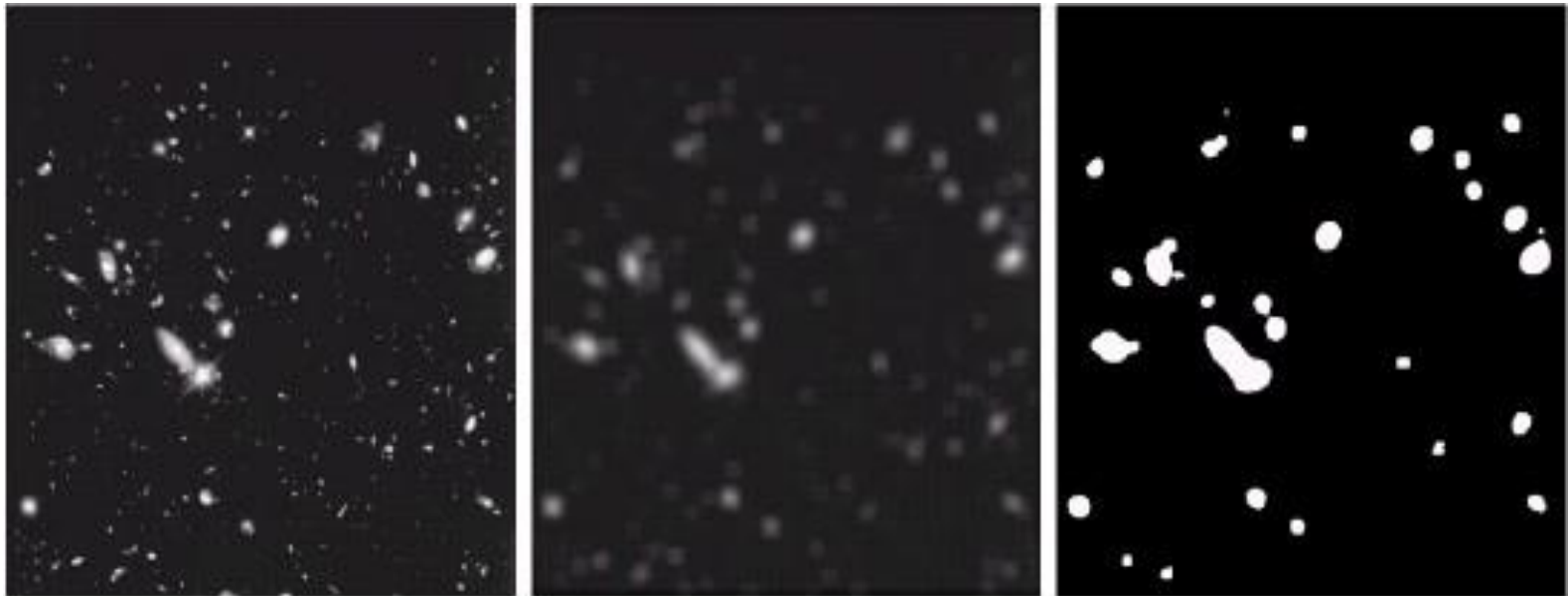


Smoothed by
35 x 35 box filter



Spatial Filtering for Smoothing:: Example

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a b c

FIGURE 3.36 (a) Image from the Hubble Space Telescope. (b) Image processed by a 15×15 averaging mask. (c) Result of thresholding (b). (Original image courtesy of NASA.)

Weighted Smoothing Spatial Filters

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- More effective smoothing filters can be generated by allowing different pixels in the neighbourhood different weights in the averaging function
 - ✓ Pixels **closer** to the central pixel are more important
 - ✓ Often referred to as *a weighted averaging*

$\frac{1}{16}$	$\frac{2}{16}$	$\frac{1}{16}$
$\frac{2}{16}$	$\frac{4}{16}$	$\frac{2}{16}$
$\frac{1}{16}$	$\frac{2}{16}$	$\frac{1}{16}$

Weighted Averaging Filter

Order Statistics Filtering

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- ❑ Nonlinear spatial filters
- ❑ Output is based on order of gray levels in the masked area (sub-image)
- ❑ Examples: Median filtering, Max & Min filtering
 - ✓ **Min:** Set the pixel value to the minimum in the neighbourhood
 - ✓ **Max:** Set the pixel value to the maximum in the neighbourhood
 - ✓ **Median:** The median value of a set of numbers is the midpoint value in that set (e.g. from the set [1, 7, 15, 18, 24] 15 is the median). Sometimes the median works better than the average

Order Statistics Filtering

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Median Filtering

- Assigns the mid value of all the gray levels in the mask to the center of mask;
- Particularly effective when
 - ✓ *the noise pattern consists of strong, spiky components (**salt-and-pepper**)*
 - ✓ *edges are to be preserved*
 - ✓ *Force points with distinct gray levels to be more like their neighbors*

10	20	20
20	15	20
20	25	100



Output = ? **20**

Order Statistics Filtering

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Minimum Filtering

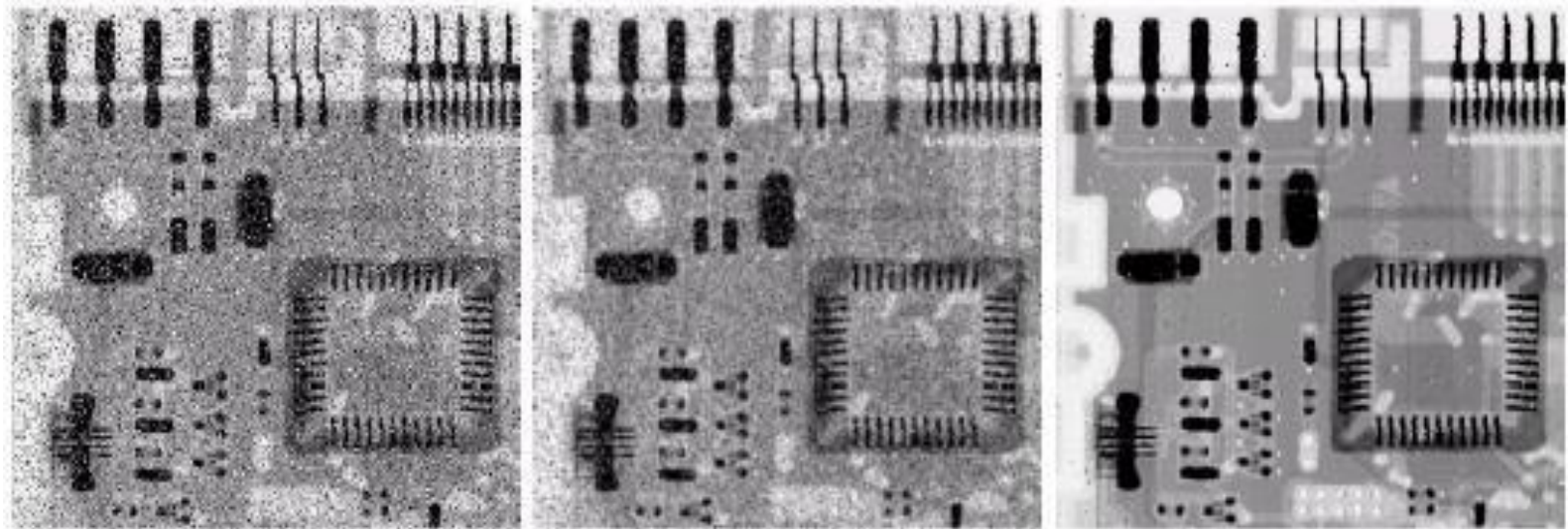
- Assigns the minimum value of all the gray levels in the mask to the center of mask;
- Particularly effective when
 - ✓ *Only Salt noise is present*

Maximum Filtering

- Assigns the maximum value of all the gray levels in the mask to the center of mask;
- Particularly effective when
 - ✓ *Only Pepper noise is present*
- *Edges are to be preserved*
- *Force points with distinct gray levels to be more like their neighbors*

Order Statistics Filtering

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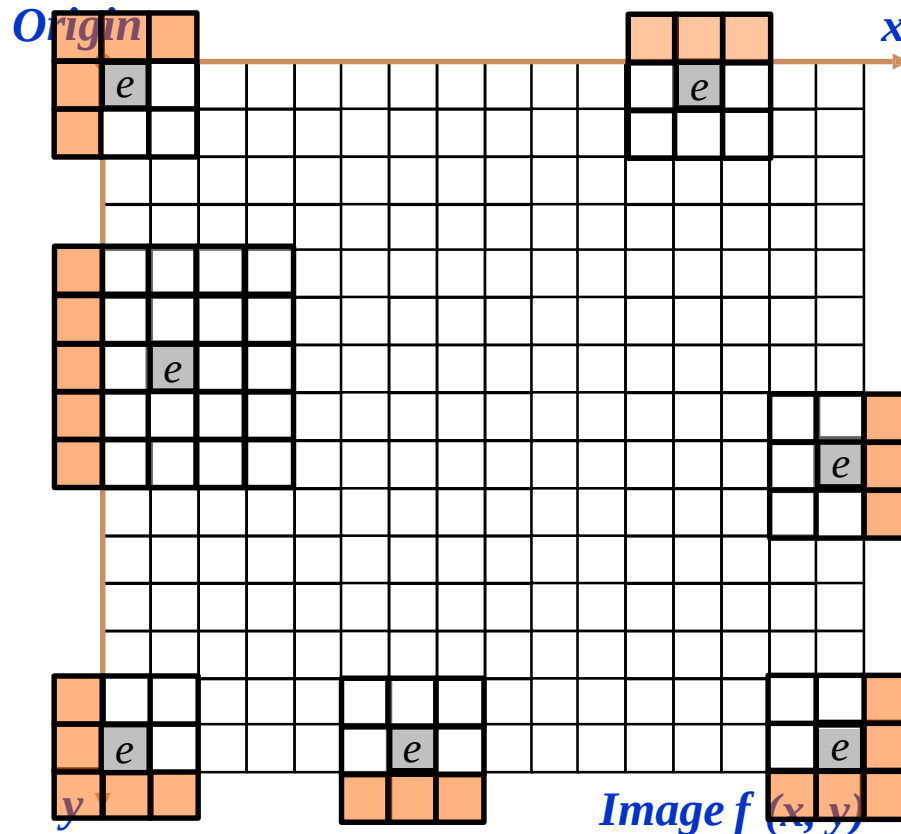
a b c

FIGURE 3.37 (a) X-ray image of circuit board corrupted by salt-and-pepper noise. (b) Noise reduction with a 3×3 averaging mask. (c) Noise reduction with a 3×3 median filter. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

Strange Things Happen At The Edges!

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- At the edges of an image we are missing pixels to form a neighbourhood



Border Pixels Treatment

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- ❑ Mask operation near the image border
- ❑ Problem arises when part of the mask is located outside the image plane; to handle the problem:
 - ✓ Truncate the image
 - Discard the problem pixels (e.g. 512x512 input 510x510 output, if mask size is 3x3)
 - ✓ Image Padding
 - expand the input image by padding zeros or 255(512x512 input 514x514output)
 - *Image padding is not good; creates artificial lines or edges on the border*

Border Pixels Treatment

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Replicate border pixels

- We normally use the gray levels of border pixels to fill up the expanded region (for 3x3 mask). For larger masks a border region equal to half of the mask size is mirrored on the expanded region.

102	102	130	143	123	115	...	...
102	102	130	143	123	115	...	...
93	93	...	...	...			
98	98	...					
82	82	...					
65	65						
...	...						
...	...						

Expanded area

Original image size (shaded area)

Histogram

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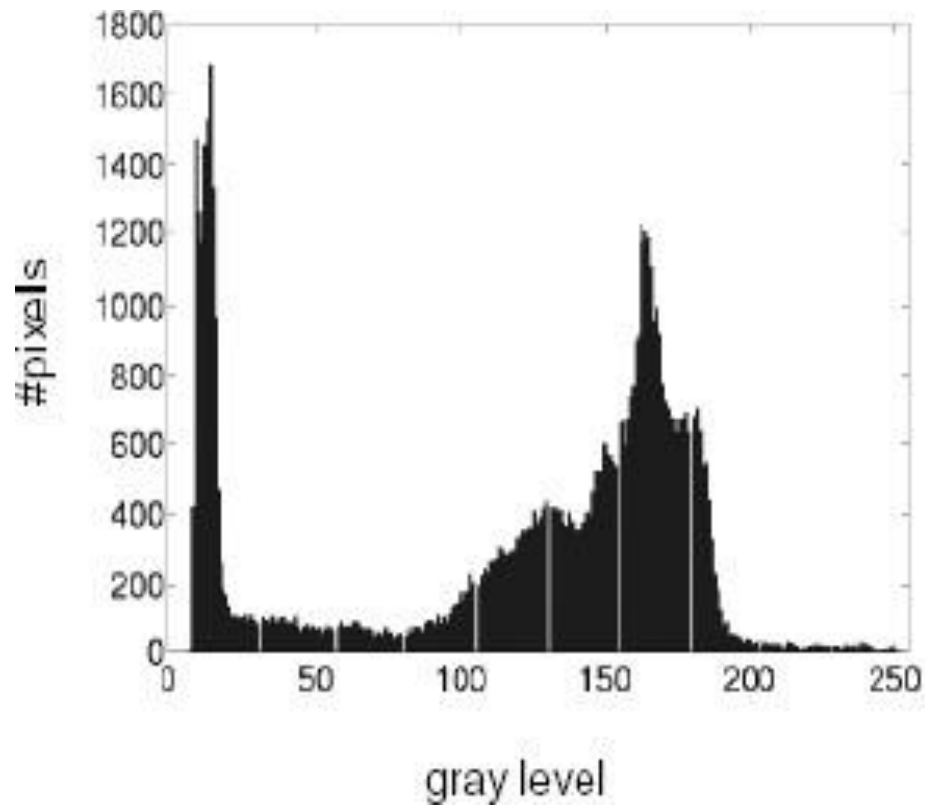
- The histogram of an image represents the frequency of pixels in an image
- The histogram of an image shows the distribution of gray levels in the image
- Massively useful in image processing especially in image segmentation
- Histogram of an image with gray level (0 to $L-1$):

A discrete function $h(r_k) = n_k$, where r_k is the k^{th} gray level and n_k is the number of pixels in the image having gray level r_k .

- How to obtain histogram?
 - ✓ For B bit image, initialize 2^B counters with 0
 - ✓ Loop over all pixels x, y
 - ✓ When encountering gray level $f(x, y) = I$, increment counter # i

Histogram : Example

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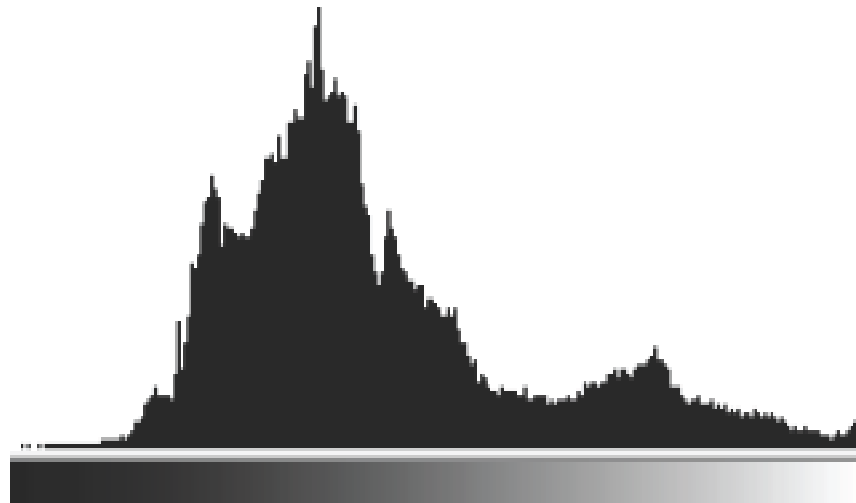


Cameraman
image

Histogram and Image Brightness/Contrast

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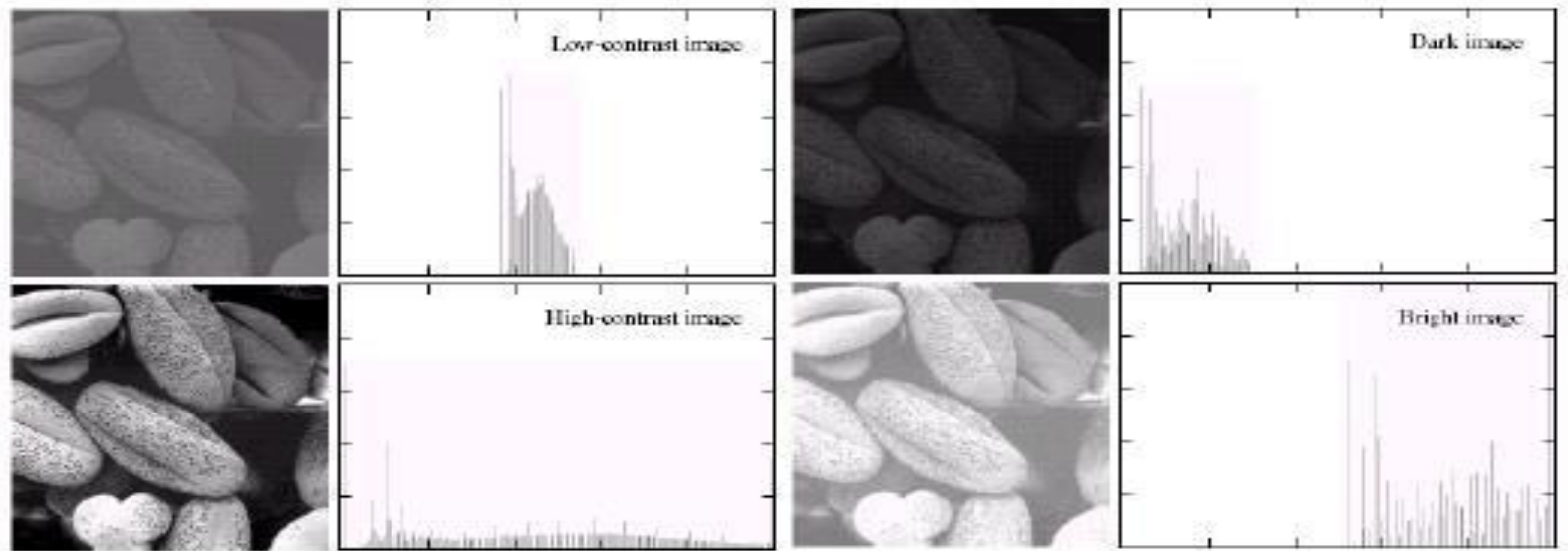
- Changing the size of the image shrinks or stretches the histogram vertically.
- Adjusting brightness shifts the histogram horizontally.
- Adjusting contrast causes the histogram to stretch or shrink horizontally:
 - ✓ **Stretching = more contrast**
 - ✓ **Shrinking = less contrast**



Histogram : Example

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- A selection of images and their Histograms
- Note that the high contrast image has the most evenly spaced histogram
- Histograms of low contrast images are located in certain portions and not in the entire gray scale range

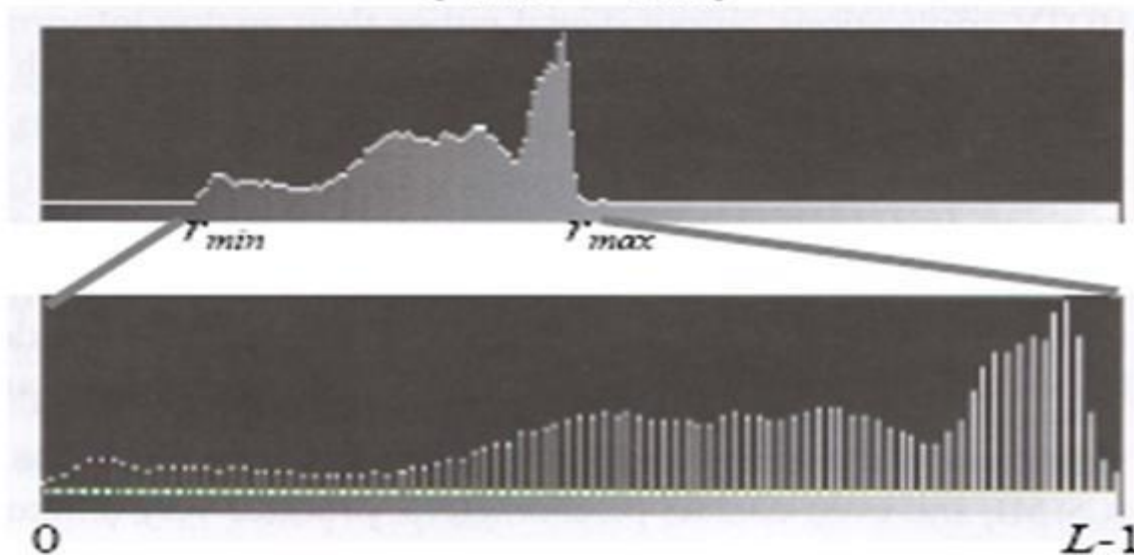


Contrast Stretching through Histogram

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- If r_{max} and r_{min} are the maximum and minimum gray level of the input image and L is the total gray levels of output image The transformation function for contrast stretching will be

$$s = T(r) = (r - r_{min}) \left(\frac{L - 1}{r_{max} - r_{min}} \right)$$



Histogram Equalization

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- The idea is to find a transformation $s=T(r)$ to be applied to each pixel of the input image $f(x,y)$ such that a uniform distribution of gray levels in the entire range results for the output image $g(x,y)$

The discrete approximation of the transformation function for histogram equalization is:

$$s_k = T(r_k) = \sum_{j=0}^k p_r(r_j) \quad \text{for } 0 \leq k \leq L-1$$

where $p_r(r_j) = \frac{n_j}{n}$, $j = 0, \dots, L-1$ and $n = \sum_{j=0}^{L-1} n_j$

Histogram Equalization

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- ❑ Spreading out the frequencies in an image (or equalising the image) is a simple way to improve dark or washed out images
- ❑ The formula for histogram equalization is given where
 - ✓ r_k : input intensity
 - ✓ s_k : processed intensity
 - ✓ k : the intensity (e.g 0.0 – 1.0)
 - ✓ n_j : the frequency of intensity j
 - ✓ n : the sum of all frequencies (total pixels)

$$\begin{aligned} s_k &= T(r_k) \\ &= \sum_{j=1}^k p_r(r_j) \\ &= \sum_{j=1}^k \frac{n_j}{n} \end{aligned}$$

Histogram Equalization

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52	55	61	66	70	61	64	73
63	59	55	90	109	85	69	72
62	59	68	113	144	104	66	73
63	58	71	122	154	106	70	69
67	61	68	104	126	88	68	70
79	65	60	70	77	68	58	75
85	71	64	59	55	61	65	83
87	79	69	68	65	76	78	94

Value	Count	Value	Count	Value	Count	Value	Count	Value	Count
52	1	64	2	72	1	85	2	113	1
55	3	65	3	73	2	87	1	122	1
58	2	66	2	75	1	88	1	126	1
59	3	67	1	76	1	90	1	144	1
60	1	68	5	77	1	94	1	154	1
61	4	69	3	78	1	104	2		
62	1	70	4	79	2	106	1		
63	2	71	2	83	1	109	1		

Histogram Equalization

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52	55	61	66	70	61	64	73
63	59	55	90	109	85	69	72
62	59	68	113	144	104	66	73
63	58	71	122	154	106	70	69
67	61	68	104	126	88	68	70
79	65	60	70	77	68	58	75
85	71	64	59	55	61	65	83
87	79	69	68	65	76	78	94

Initial Image

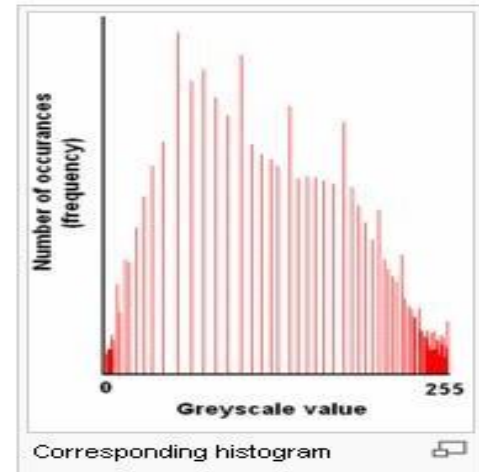
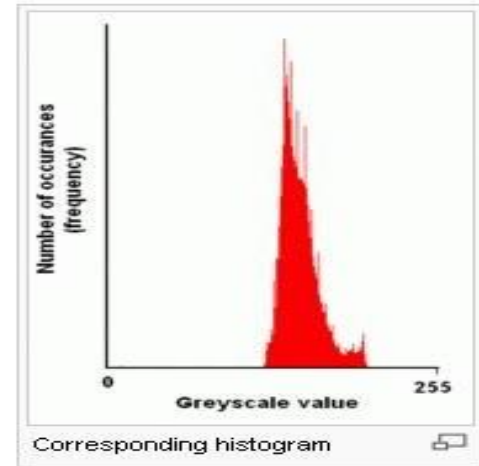
0	12	53	93	146	53	73	166
65	32	12	215	235	202	130	158
57	32	117	239	251	227	93	166
65	20	154	243	255	231	146	130
97	53	117	227	247	210	117	146
190	85	36	146	178	117	20	170
202	154	73	32	12	53	85	194
206	190	130	117	85	174	182	219

Image After Equalization

Notice that the minimum value (52) is now 0 and the maximum value (154) is now 255.

Histogram Equalization

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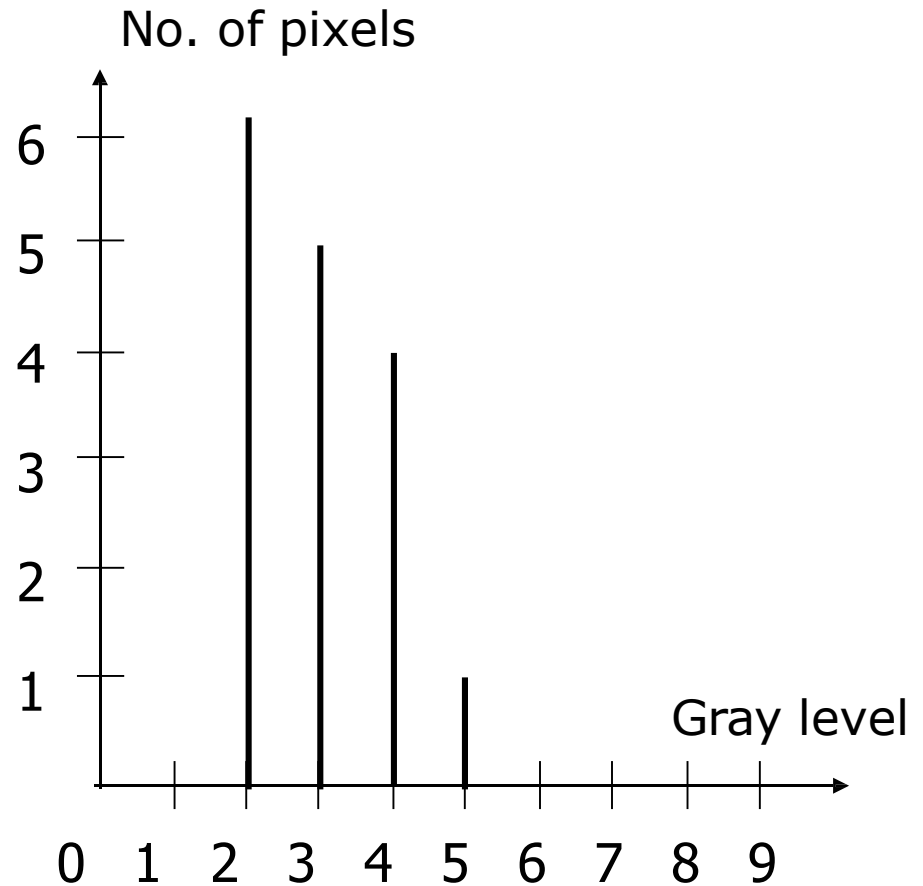
Histogram Equalization

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2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4

4x4 image

Gray scale = [0,9]



Histogram Equalization

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Gray Level(j)	0	1	2	3	4	5	6	7	8	9
No. of pixels	0	0	6	5	4	1	0	0	0	0
PDF <i>Probability Density Function</i>	0/16	0/16	6/16	5/16	4/16	1/16	0/16	0/16	0/16	0/16
CDF <i>Cumulative Distribution Function</i>	0	0	6/16	11/16	15/16	16/16	1	1	1	1
s x 9	0	0	3.3≈3	6.1≈6	8.4≈8	9	9	9	9	9

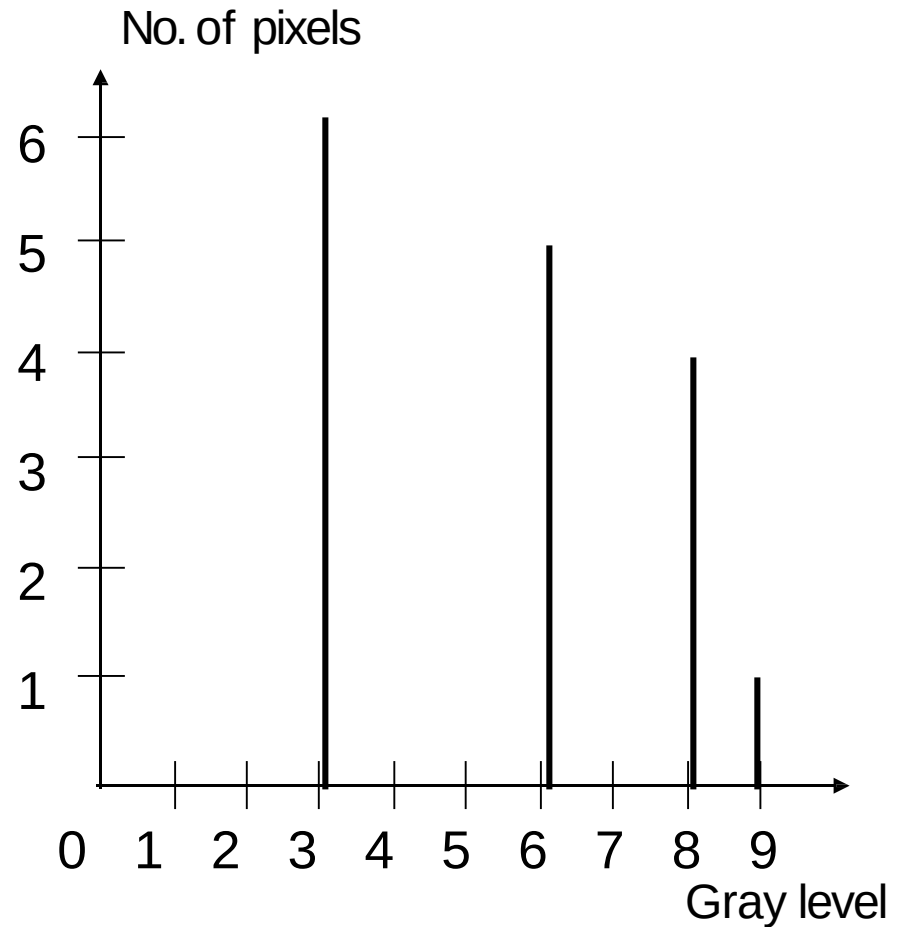
Histogram Equalization

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3	6	6	3
8	3	8	6
6	3	6	9
3	8	3	8

Output image

Gray scale = [0,9]



Equalized histogram

Local Enhancement through Histogram Processing

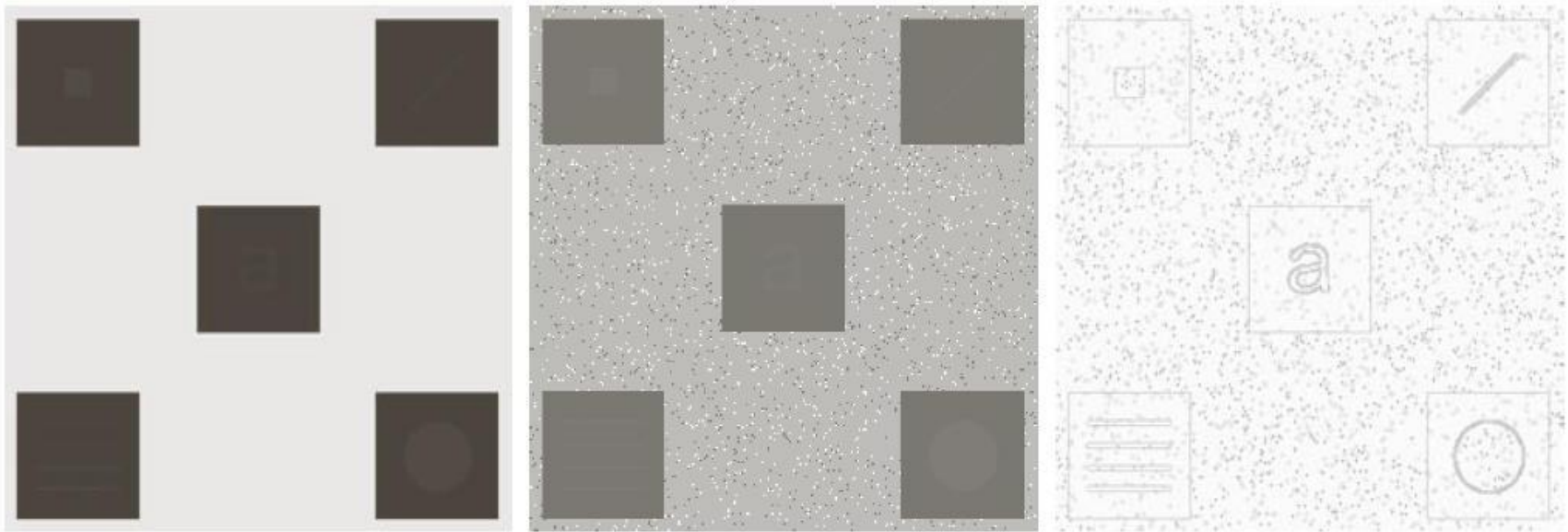
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- ❑ The histogram processing methods discussed earlier are global in the sense that pixels are modified by a transformation function based on the gray level content of the entire image
- ❑ Although this global approach is suitable for overall enhancement. There are cases in which it is necessary to enhance details over small areas in the image
- ❑ The histogram equalization can be easily adapted to local enhancement
- ❑ The procedure is to define a neighborhood and move the center of this area from pixel to pixel
- ❑ At each location, the histogram of the points in the neighborhood is computed and then histogram transformation function is obtained

Local Enhancement through Histogram Processing

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- Map the intensity of the pixel centered in the neighborhood
- Move to the next location and repeat the procedure



a b c

FIGURE 3.26 (a) Original image. (b) Result of global histogram equalization. (c) Result of local histogram equalization applied to (a), using a neighborhood of size 3×3 .

Histogram Equalization Exercise

Gray scale = $[0,9]$

4	3	6	2
3	2	4	1
4	6	2	0
8	4	1	5

Histogram Equalization Exercise Solution

7	5	8	3
5	3	7	2
7	8	3	1
9	7	2	7

References

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1. DIP by Gonzalez

For any query Feel Free to ask



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